Li 2019 melanoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Harmony reaches near-maximal sample mixing (iLISI 1.00→1.78 of 3), but the malignant structure resolves into TWO density lobes that are transcriptomically disconnected — inter-cluster PAGA on the PCA-space graph ≈ 0.001 (~0), invariant across n_pcs 20/30 × θ 10/15, and stronger integration never merges them. Neither lobe reproduces the embryonic axis (in both, Gast/pSom ANTI-correlate rather than co-vary). A broken manifold with no cell-state continuum — a trajectory is not forced.

Gate 2 — module geography → N/A

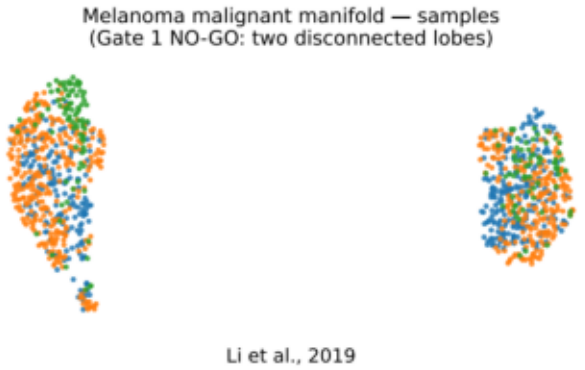
Not evaluated — Gate 1 NO-GO halts the chain before the signature step.

Gate 3 — cross-repression → N/A

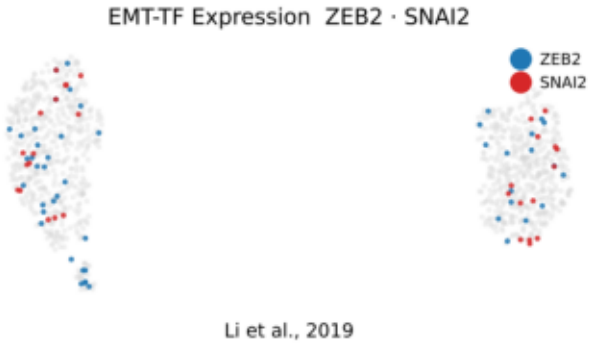
Not evaluated — branching-only; no trajectory fit.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Melanoma (Li et al., 2019), MARS-seq immunotherapy atlas — 1,082 malignant cells across 3 tumours (3/20 clear the ≥56-cell threshold). Gate 1 NO-GO: two disconnected lobes.

Verdict.
The embryonic-program pattern does NOT reproduce in Li et al. 2019. The small malignant compartment (1,082 cells, 3 tumours) forms two transcriptomically disconnected lobes rather than one integrated, continuous cell-state manifold, and the Gast/pSom/Crest modules are salt-and-pepper with no coherent geography — Gast and pSom anti-correlate in each lobe instead of co-varying. Gate 1 is a NO-GO; Gates 2–3 are not evaluated. EMT-TFs (ZEB2, SNAI2) are reported unconditionally on the best-attempt embedding, without a pseudotime claim.

Parameter choices

Malignant load	1,160 cells of 78,336 (MARS-seq atlas); Malignant subset only	**Sample threshold**	≥56 cells (largest gap 154→21, ratio 7.3) → p25/p21/p26, 1,082 cells
Overdispersed genes	217 (scFates GAM); PCA n_pcs=20	**Integration**	Harmony on sample θ=10 (near-maximal iLISI; higher over-corrects, no merge)
Continuity test	UMAP radius-graph lobes + PAGA on the PCA graph (layout-independent)	**EMT-TFs**	ZEB2 (43%) + SNAI2 (27%), top-decile expressers; no pseudotime claim
Geometry	none — trajectory not fit (Gate 1 NO-GO)		